

Configuration-Driven Architecture Search Space

Unified configuration and sweep interface
Config + YAML sweeps



Representation

- hidden-irrep strings
- multiplicities and $l_{\max}$
- parity channels
- IrrepsAutoBuilder,
build_hidden_irreps()

Encoders

- NodeEncoder
- EdgeEncoder
- radial scaling and SH preprocessing
- edge-encoder style selection

Message passing

- layer depth
- node and edge residuals
- self-connections
- EquiConv, MessageBlock

Tensor products and nonlinearities

- separate-weight / fully connected TP
- NormAct, Gate
- GateScalarsMLP, GateMagnitudes
- extended E3-MLP variants

Readout heads

- head depth
- shifted-self branches
- tensor-square options
- scalar and magnitude variants
- DeepHead

Objectives

- target subset selection
- observable guidance terms
- force and stress weights
- optimizer and scheduler controls
- E3GNN._shared_step()